Gurugram University Scheme of Studies and Examination
Bachelor of Technology Semester 6

S.No.	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total
				L	T	P				
1	PCC		Power System II	3	1	0	3	30	70	100
2	PCC		Control System	3	1	0	3	30	70	100
3	PCC		Microprocessors and Microcontrollers	3	1	0	3	30	70	100
4	PEC		Professional Electives II	3	0	0	3	30	70	100
5	PEC		Professional Electives III	3	0	0	3	30	70	100
6	OEC		Open Elective II	3	0	0	3	30	70	100
7	LC		Power System II Lab	0	0	2	1	50	50	100
8	LC		Control System Lab	0	0	2	1	50	50	100
9	LC		Microprocessors and Microcontrollers Lab	0	0	2	1	50	50	100
10	PROJ		Project-I	0	0	4	2	50	50	100
11	HSMC		Economics for Engineers*	2	0	0	-	30	70	100*
Total							**23**			**1000**

NOTE:
1. At the end of the 6th semester, each student has to undergo Practical Training of 4/6 weeks in an Industry/ Institute/ Professional Organization/ Research Laboratory/ training center etc. and submit the typed report along with a certificate from the organization and its evaluation shall be carried out in the 7th Semester.
2. Choose any one from each of the Professional Elective Course-II and III
3. Choose any one from Open Elective Course-II

PROFESSIONAL ELECTIVE- II (Semester-VI)

Sr. No	Code	Subject	Credit
1.		Robotics and Automation	3
2.		Energy Management and Auditing	3
3.		Introduction to MEMS	3
4.		Wireless Sensor Networks	3
5.		Mobile Communications	3

PROFESSIONAL ELECTIVE- III (Semester-VI)

Sr. No	Code	Subject	Credit
1.		Power Plant Engineering	3
2.		Power System Protection	3
3.		Electrical and Hybrid Vehicle	3
4.		Modelling and Analysis of Electrical Machines	3
5.		Electrical Safety and Standards	3

***Note:** The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

POWER SYSTEM-II

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power System II</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. To learn about power flow analysis.
2. To gain understanding of economics of power system.
3. To understand voltage and load frequency control.
4. To learn about power system stability.

Unit-I

Power Flow Analysis: Review of the structure of a Power System and its components. Analysis of Power Flows: Formation of Bus Admittance Matrix. Real and reactive power balance equations at a node. Application of numerical methods for solution of nonlinear algebraic equations – Gauss Seidel and Newton-Raphson methods for the solution of the power flow equations.

Unit-II

Economic Operation of Power Systems: Distribution of loads between units within a plant. Distribution of loads between plants, Transmission loss equation, Classical Economic dispatch with losses. Optimal unit commitment problems and their solutions.

Unit-III

Voltage and Load Frequency Control: Introduction to control of active and reactive power flow, control of voltage, Excitation systems. Introduction to Load Frequency Control and Automatic generation control, Single area and modelling of AGC, Concept of multi area AGC.

Unit-IV

Power System Stability: Concepts, steady state and transient stability, swing equations, equal area criterion. Solution of Swing Equation, Transient stability algorithm using modified Euler's method and fourth order Runga Kutta method, – multi-machine stability analysis.

Course Outcomes:

At the end of this course, students will demonstrate the ability to;
1. Use numerical methods to analyse a power system in steady state.
2. Formulate Ybus and Zbus.
3. Apply load flow analysis on a power system.
4. Understand stability constraints in a synchronous grid.
5. Understand methods to control the voltage, frequency and power flow.
6. Understand the basics of power system economics

Text/References:

1. J. Grainger and W. D. Stevenson, "Power System Analysis", McGraw Hill Education, 1994.
2. O. I. Elgerd, "Electric Energy Systems Theory", McGraw Hill Education, 1995.
3. A. R. Bergen and V. Vittal, "Power System Analysis", Pearson Education Inc., 1999.
4. D. P. Kothari and I. J. Nagrath, "Modern Power System Analysis", McGraw Hill Education, 2003
5. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake and G. Strbac, "Electric Power Systems", Wiley, 2012.

CONTROL SYSTEM

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Control System</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand concepts of the mathematical modeling, feedback control and stability analysis in Time and Frequency domains
2. To develop skills, to analyze feedback control systems in continuous- and discrete time domains.
3. To learn methods for improving system response transient and steady state behavior (response).
4. The compensator design of linear systems is also introduced.

Unit-I

Systems Components and Their Representation Control System: Terminology and Basic Structure-Feed forward and Feedback control theory-Electrical and Mechanical Transfer Function Models-Block diagram Models-Signal flow graphs models-DC and AC servo Systems-Synchronous -Multivariable control system

Unit-II

Time Response Analysis and Stability Concept Transient response-steady state response-Measures of performance of the standard first order and second order system-effect on an additional zero and an additional pole-steady error constant and system- type number-PID control.

Concept of stability-Bounded - Input Bounded - Output stability-Routh stability criterion- Relative stability-Root locus concept-Guidelines for sketching root locus.

Unit-III

Frequency Domain Analysis Bode Plot - Polar Plot- Nyquist Plots-Design of compensators using Bode Plots-Cascade lead compensation-Cascade lag compensation-Cascade lag-lead compensation

Unit-IV

Control System Analysis Using State Variable Methods State variable representation-Conversion of state variable models to transfer functions-Conversion of transfer functions to state variable models-Solution of state equations-Concepts of Controllability and Observability-Stability of linear systems-Equivalence between transfer function and state variable representations

Course Outcomes: At the end of this course students will demonstrate the ability to

1. Understand the concepts of control systems and importance of feedback in control systems.
2. Perform signal flow graph and formulate transfer function.
3. Perform computations and solve problems on frequency response analysis.
4. Analyse Polar, Bode and Nyquist's plot.
5. Evaluate different types of state models and time functions.
6. Analyse different types of control systems like linear and non-linear control systems, etc.

Text/Reference Books:

2. Gopal. M., "Control Systems: Principles and Design", Tata McGraw-Hill, 1997
3. Ambikapathy A., Control Systems, Khanna Book Publications, 2019.
4. Kuo, B.C., "Automatic Control System", Prentice Hall, sixth edition, 1993.
5. Ogata, K., "Modern Control Engineering", Prentice Hall, second edition, 1991.
6. Nagrath and Gopal, "Modern Control Engineering", New Age International, New Delhi

MICROPROCESSORS AND MICROCONTROLLERS

Course Code					
Category	**Professional Core Courses**				
Course title	**Microprocessors and Microcontrollers**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: VI**
	3	**1**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective:
The objectives of this course.

1. To develop an in-depth understanding of the operation of microprocessors and microcontroller.
2. To master the assembly language programming using concepts like assembler directives, procedures, macros, software interrupts etc.
3. To create an exposure to basic peripherals, its programming and interfacing techniques.
4. To understand the concept of Interrupts and interfacing details of 8086 and microcontroller.

Unit-I

THE 8086 MICROPROCESSOR
Introduction to 8086 — Microprocessor architecture — Addressing modes — Instruction set and assembler directives — Assembly language programming — Modular Programming — Linking and Relocation — Stacks — Procedures — Macros — Interrupts and interrupt service routines — Byte and String Manipulation.

Unit-II

8086 SYSTEM BUS STRUCTURE
8086 signals — Basic configurations — System bus timing –System design using 8086 — I/O programming — Introduction to Multiprogramming — System Bus Structure — Multiprocessor configurations — Coprocessor, closely coupled and loosely Coupled configurations — Introduction to advanced processors.

Unit-III

I/O INTERFACING
Memory Interfacing and I/O interfacing — Parallel communication interface — Serial communication interface — D/A and A/D Interface — Timer — Keyboard /display controller — Interrupt controller — DMA controller — Programming and applications Case studies: Traffic Light control, LED display, LCD display, Keyboard display interface and Alarm Controller.

Unit-IV

MICROCONTROLLER
Architecture of 8051 — Special Function Registers (SFRs) — I/O Pins Ports and Circuits — Instruction set — Addressing modes — Assembly language programming.
INTERFACING MICROCONTROLLER
Programming 8051 Timers — Serial Port Programming — Interrupts Programming — LCD and Keyboard Interfacing — ADC, DAC and Sensor Interfacing — External Memory Interface- Stepper Motor and Waveform generation — Comparison of Microprocessor, Microcontroller, PIC and ARM processors

Course Outcomes:
At the end of this course students will be able to:

1. Understand the fundamentals of Microprocessors.
2. Understand the internal design of 8051 microcontroller along with the features and their programming.
3. Competent with the on-chip peripherals of microcontrollers.
4. Design different interfacing applications using microcontrollers and peripherals.
5. Demonstrate the limitations and strengths of different types of microcontrollers and their comparison.
6. Build systems using microcontrollers for real time applications.

List of References:

1. Ramesh S. Gaonkar, “Microprocessor Architecture, Programming, and Applications with the 8085”, 5th Edition, Penram International, 2009.
2. Douglas Hall, “Microprocessor and Interfacing”, 2nd Edition, TMH, 2006.
3. Muhammad A. Mazidi, “The 8051 Microcontroller And Embedded Systems Using Assembly and C”, 2nd Edition., PHI, 2012.
4. Kenneth J. Ayala, “The 8051 Microcontroller”, 3rd Edition., Cengage Learning Publication, 2007.
5. Ajit Pal, “Microcontrollers: Principals and Applications”, 2nd Edition, PHI, 2011. 6. Datasheet of P89V51RD2

ROBOTICS AND AUTOMATION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Robotics and Automation</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
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<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Course Objectives:
1. To learn about relationship between mechanical structures of industrial robots.
2. To gain understanding of spatial transformation to obtain forward kinematic equation of robot manipulators.
3. To understand inverse kinematics of simple robot manipulators.
4. To learn about Jacobian matrix and use it to identify singularities.

Unit-I
Introduction: Concept and scope of automation: Socio economic impacts of automation, Types of Automation, Low-Cost Automation
Fluid Power: Fluid power control elements, Standard graphical symbols, Fluid power generators, Hydraulic and pneumatic Cylinders - construction, design and mounting; Hydraulic and pneumatic Valves for pressure, flow and direction control.

Unit-II
Basic hydraulic and pneumatic circuits: Direct and Indirect Control of Single/Double Acting Cylinders, designing of logic circuits for a given time displacement diagram and sequence of operations, Hydraulic and Pneumatic Circuits using Time Delay Valve and Quick Exhaust Valve, Memory Circuit and Speed Control of a Cylinder, Troubleshooting and "Causes and Effects of Malfunctions" Basics of Control Chain, Circuit Layouts, Designation of specific Elements in a Circuit.
Fluidics: Boolean algebra, Truth Tables, Logic Gates, Coanda effect.

Unit-III
Electrical and Electronic Controls: Basics of Programmable logic controllers (PLC), Architecture and Components of PLC, Ladder Logic Diagrams
Transfer Devices and feeders: Classification, Constructional details and Applications of Transfer devices, Vibratory bowl feeders, Reciprocating tube, Centrifugal hopper feeders

Unit-IV
Robotics: Introduction, Classification based on geometry, control and path movement, Robot Specifications, Robot Performance Parameters, Robot Programming, Machine Vision, Teach pendants, Industrial Applications of Robots

Course Outcomes (COs): After studying this course, students will be able:
1. to demonstrate knowledge of the relationship between mechanical structures of industrial robots and
2. to learn robot's operational workspace characteristics.
3. to demonstrate an ability to apply spatial transformation to obtain forward kinematic equation of robot manipulators.
4. to learn PLC
5. to demonstrate an ability to solve inverse kinematics of simple robot manipulators.
6. to demonstrate an ability to obtain the Jacobian matrix and use it to identify singularities.

Text Books:

1. Anthony Esposito, Fluid Power with applications, Pearson
2. S. R Majumdar, Pneumatic Control, McGraw Hill
3. S. R Deb, Robotic Technology and Flexible Automation, Tata Mc Hill
4. Saeed B. Niku Introduction to Robotics, Wiley India
5. Ashitava Ghosal, Robotics, Oxford

ENERGY MANAGEMENT AND AUDITING

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Energy Management and Auditing</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Course Objectives:
1. To illustrate the concept energy management.
2. To introduce to energy audit study.
3. To study the basics of electrical energy management.
4. To learn about alternative energy.

Unit-I

Introduction: Introduction to energy management, Organizational Structure, Energy Policy and planning, Peak Demand controls, Methodologies, Types of Industrial Loads, Optimal Load Scheduling-Case studies.

Unit-II

Energy Auditing: Introduction, Energy Auditing Services, Basic Components of an Energy Audit, Specialized Audit Tools, Industrial Audits, Commercial Audits, Residential Audits, Indoor Air Quality and basics of economic analysis, cash flow model, time value of money, evaluation of proposals, pay-back method, average rate of return method, internal rate of return method, present value method, life cycle costing approach, Case studies.

Unit-III

Electric Energy Management: Introduction, Power Supply Effects of Unbalanced Voltages on the Performance of Motors, Power Factor, Electric motor Operating Loads, Determining Electric Motor Operating Loads, Power Meter, Slip Measurement, Electric Motor Efficiency, Sensitivity of Load to Motor RPM, Theoretical Power Consumption, Motor Efficiency Management, Motor Performance Management Process

Unit-IV

Alternative Energy: Introduction, Solar Energy, Wind Energy and other renewable resources for energy management.

Course Outcomes (COs): After studying this course, students will be able:
1. Understand the fundamentals of energy management systems.
2. Carry out various energy audit processes.
3. Describe Indoor Air Quality and basics of economic analysis
4. Understand various factors affecting performance of system.
5. Describe methods to improve efficiency of electrical energy systems.
6. Asses the use of alternative energy sources in improving the energy management

Text Books:

1. Wayne C. Turner, Steve Doty, Energy Management Handbook, The Fairmont Press, Inc.
2. Barney L. Capehart, Wayne C. Turner, William J. Kennedy, Guide to Energy Management, CRC Press.
3. Albert Thumann, William J. Younger, Handbook of Energy Audits, CRC Press, 2003.
4. Charles M. Gottschalk, Industrial energy conservation, John Wiley & Sons, 1996.
5. Craig B. Smith, Energy management principles, Pergamon Press.
6. D. Yogi Goswami, Frank Kreith, Energy Management and Conservation Handbook, CRC Press, 2007
7. G.G. Rajan, Optimizing energy efficiencies in industry -, Tata McGraw Hill, Pub. Co., 2001.
8. IEEE recommended practice for energy management in industrial and commercial facilities, IEEE std 739 - 1995
9. M Jayaraju and Premlet, Introduction to Energy Conservation and Management, Phasor Books, 2008
10. Paul O’Callaghan, Energy management, McGraw Hill Book Co.

INTRODUCTION TO MEMS

Course Code					
Category	**Professional Elective Courses**				
Course title	**Introduction to MEMS**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: VI**
	3	**0**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

1. To illustrate the concept of micro/nanosystems.
2. To introduce to state-of-the-art lithography techniques for micro/nanosystems.
3. To study the new materials, science and technology for micro/nanosystem applications
4. To learn about state-of-the-art micromachining and packaging technologies.

Unit-I

Overview of MEMS and Microsystems: Introduction Microsystems vs. MEMS, Microsystems and Microelectronics, the Multidisciplinary Nature of Microsystems design and manufacture, Application of MEMS in various industries. MEMS and Miniaturization: Scaling laws in miniaturization: Introduction to Scaling, Scaling in Geometry, Rigid Body dynamics, Electrostatic forces, Electromagnetic forces, Electricity, Fluid Mechanics, Heat Transfer, Over view of Micro/Nano Sensors, Actuators and Systems.

Unit-II

Review of Basic MEMS fabrication modules: Oxidation, Deposition Techniques, Lithography (LIGA), and Etching. Micromachining: Surface Micromachining, sacrificial layer processes, Stiction; Bulk Micromachining, Isot ropic Etching and Anisotropic Etching, Wafer Bonding.

Unit-III

Mechanics of solids in MEMS/NEMS: Stresses, Strain, Hookes's law, Poisson effect, Linear Thermal Expansion, Bending; Energy methods.

Unit-IV

Overview of Finite Element Method, Modeling of Coupled Electromechanical Systems: electrostatics, coupled electro mechanics.

Course Outcomes (COs): After studying this course, students will be able:

1. Be introduced to the field of micro/nanosystems
2. Grain a knowledge of basic approaches for micro/nanosystem design
3. Gain a knowledge of state-of-the-art lithography techniques for micro/nanosystems.
4. Learn new materials, science and technology for micro/nanosystem applications.
5. Understand materials science for micro/nanosystem applications
6. Understand state-of-the-art micromachining and packaging technologies

Text Books:

1. G. K. Ananthasuresh, K. J. Vinoy, S. Gopalkrishnan K. N. Bhat, V. K. Aatre, Micro and Smart Systems, Wiley India, 2012.
2. S. E. Lyshevski, Nano-and Micro-Electromechanical systems: Fundamentals of Nano and Microengineering (Vol. 8). CRC press, (2005).
3. S. D. Senturia, Microsystem Design, Kluwer Academic Publishers, 2001.
4. M. Madou, Fundamentals of Microfabrication, CRC Press, 1997.
5. G. Kovacs, Micromachined Transducers Sourcebook, McGraw-Hill, Boston, 1998.
6. M.H. Bao, Micromechanical Transducers: Pressure sensors, accelerometers, and Gyroscopes, Elsevier, New York, 2000.

WIRELESS SENSOR NETWORKS

Course Code						
Category	**Professional Elective Courses**					
Course title	**Wireless Sensor Networks**					
Scheme	**L**	**T**	**P**	**Credits**	**Semester: VI**	
	3	**0**	**0**	**3**		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	3Hrs					

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. Understand the working principles of the Sensors.
2. Understand the protocols used in sensor networks.
3. Understand design principles of WSN.
4. Understand engineering sensor networks.

Unit-I

Introduction to Sensor Networks, unique constraints and challenges, Advantage of Sensor Networks, Applications of Sensor Networks, Types of wireless sensor networks, Mobile Adhoc Networks (MANETs) and Wireless Sensor Networks, Enabling technologies for Wireless Sensor Networks. Issues and challenges in wireless sensor networks.

Unit-II

Routing protocols, MAC protocols: Classification of MAC Protocols, S-MAC Protocol, BMAC protocol, IEEE 802.15.4 standard and ZigBee

Unit-III

Dissemination protocol for large sensor network. Data dissemination, data gathering, and data fusion; Quality of a sensor network; Real-time traffic support and security protocols. Design Principles for WSNs, Gateway Concepts Need for gateway, WSN to Internet Communication, and Internet to WSN Communication

Unit-IV

Single-node architecture, Hardware components and design constraints, Operating systems and execution environments, introduction to TinyOS and nesC

Course Outcomes (COs): After studying this course, students will be able:
1. Design wireless sensor networks for a given application
2. Understand emerging research areas in the field of sensor networks
3. Understand MAC protocols used for different communication standards used in WSN
4. Understand large sensor network.
5. Understand architecture and hardware components.
6. Explore new protocols for WSN

Text Books:

1. Waltenegus Dargie , Christian Poellabauer, “ Fundamentals Of Wireless Sensor Networks
2. Theory And Practice”, By John Wiley & Sons Publications ,2011
3. Sabrie Soloman, “Sensors Handbook" by McGraw Hill publication. 2009
4. Feng Zhao, Leonidas Guibas, “Wireless Sensor Networks”, Elsevier Publications,2004
5. Kazem Sohrby, Daniel Minoli, “Wireless Sensor Networks”: Technology, Protocols and Applications, Wiley-Inter science
5. Philip Levis, And David Gay "TinyOS Programming” by Cambridge University Press 2009

MOBILE COMMUNICATION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Mobile Communication</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
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Course Objectives:

During the course, students will be made to learn to:

1. Understand the Cellular concepts.
2. Understand the digital modulation techniques.
3. Understand the mobility in Cellular Systems.
4. Understand GSM.

Unit-I

Cellular concepts- Cell structure, frequency reuse, cell splitting, channel assignment, handoff, interference, capacity, power control; Wireless Standards: Overview of 2G and 3G cellular standards.

Unit-II

Large scale signal propagation. Fading channels-Multipath and small-scale fading- Doppler shift, doppler spread, average and rms delay spread, coherence bandwidth and coherence time, flat and frequency selective fading, slow and fast fading, average fade duration and level crossing rate.

Okumura Model, Hata Model, PCS Extension to Hata Model, Walfisch and Bertoni Model, Wideband PCS Microcell Model, Indoor Propagation Models-Partition losses (Same Floor), Partition losses between Floors, Log-distance path loss model.

Unit-III

Multiple access schemes-FDMA, TDMA, CDMA and SDMA. Modulation schemes- BPSK, QPSK and variants, QAM, MSK and GMSK, multicarrier modulation, OFDM and OFDMA.

Unit-IV

Mobility in Cellular Systems: The Gateway Concept, Measurement Reports, Mobility Procedures - Mobile IP: Basic Components, Tunneling

GSM: Architecture, – UMTS: Architecture, Basics of CDMA, - Introduction to LTE: History, Architecture - OFDM – Uplink and Downlink Communication in LTE.

Course Outcomes (COs): After studying this course, students will be able:

1. To understand the working principles of the mobile communication systems.
2. To understand the relation between the user features and underlying technology.
3. To analyze mobile communication systems for improved performance.
4. To understand multiple access schemes.
5. To analyze mobility in cellular systems.
6. To discuss GSM.

Text Books:

1. WCY Lee, Mobile Cellular Telecommunications Systems, McGraw Hill, 1990.
2. WCY Lee, Mobile Communications Design Fundamentals, Prentice Hall, 1993.
3. Raymond Steele, Mobile Radio Communications, IEEE Press, New York, 1992.
4. AJ Viterbi, CDMA: Principles of Spread Spectrum Communications, Addison Wesley, 1995.
5. VK Garg & JE Wilkes, Wireless & Personal Communication Systems, Prentice Hall, 1996.

POWER PLANT ENGINEERING

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power Plant Engineering</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

1. Acquaint the students to basic concepts of power plant.
2. To stimulate the students to think about power generation by renewable and non-renewable energy resources.
3. To acquaint the students to different types of power plant.
4. To Understand the principal components, efficiency and types of power plants.

Unit-I

Coal based thermal power plants, basic Rankine cycle and its modifications, layout of modern coal power plant, super critical boilers, FBC boilers, turbines, condensers, steam and heating rates, subsystems of thermal power plants, fuel and ash handling, draught system, feed water treatment, binary cycles and cogeneration systems

Unit-II

Gas turbine and combined cycle power plants, Brayton cycle analysis and optimization, components of gas turbine power plants, combined cycle power plants, Integrated Gasifier based Combined Cycle (IGCC) systems.

Unit-III

Basics of nuclear energy conversion, Layout and subsystems of nuclear power plants, Boiling Water Reactor (BWR), Pressurized Water Reactor (PWR), CANDU Reactor, Pressurized Heavy Water Reactor (PHWR), Fast Breeder Reactors (FBR), gas cooled and liquid metal cooled reactors, safety measures for nuclear power plants.

Unit-IV

Hydroelectric power plants, classification, typical layout and components, principles of wind, tidal, solar PV and solar thermal, geothermal, biogas and fuel cell power systems Energy, economic and environmental issues, power tariffs, load distribution parameters, load curve, capital and operating cost of different power plants, pollution control technologies including waste disposal options for coal and nuclear plants.

Course Outcomes:

Upon completion of the course students will be able to:

1. Understand the basics of Power Plants.
2. Understand the idea about the power generation by renewable and non-renewable energy resources.
3. Understand about the different types of cycles and natural resources used in power plants and their applications.
4. Understand the principal components and types of nuclear reactors.
5. Understand the principal components and types of hydro power plants.
6. Estimate different efficiencies associated with power plant systems.

Text Books:

1. Nag P.K., Power Plant Engineering, 3rd ed., Tata McGraw Hill, 2008.
2. El Wakil M.M., Power Plant Technology, Tata McGraw Hill, 2010.
3. Elliot T.C., Chen K and Swanekamp R.C., Power Plant Engineering, 2nd ed., McGraw Hill, 1998

POWER SYSTEM PROTECTION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power System Protection</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:

1. Acquaint the students to basic concepts of protection system.
2. To understand the protection schemes for different power system components.
3. To understand the basic principles of digital protection.
4. To understand system protection schemes, and the use of wide-area measurements.

Unit-I

Introduction and Components of a Protection System

Principles of Power System Protection, Relays, Instrument transformers, Circuit Breakers, Generator Protection: External and internal faults – differential protection – biased circulating current protection – self balance system – over-current and earth fault protection – protection against failure of excitation

Unit-II

Faults and Over-Current Protection

Review of Fault Analysis, Sequence Networks. Introduction to Overcurrent Protection and overcurrent relay co-ordination. Transformer protection: Differential protection – self-balance system of protection – overcurrent and earth fault protection – Buchholz's relay and its operation.

Unit-III

Equipment Protection Schemes

Directional, Distance, Differential protection. Bus bar Protection, Bus Bar arrangement schemes.

Modelling and Simulation of Protection Schemes

CT/PT modelling and standards, Simulation of transients using Electro-Magnetic Transients (EMT) programs. Relay Testing.

Unit-IV

System Protection

Effect of Power Swings on Distance Relaying. System Protection Schemes. Under-frequency, under-voltage and df/dt relays, Out-of-step protection, Synchro-phasors, Phasor Measurement Units and Wide-Area Measurement Systems (WAMS). Application of WAMS for improving protection systems.

Course Outcomes: At the end of this course, students will demonstrate the ability to

1. Understand the different components of a protection system.
2. Evaluate fault current due to different types of faults in a network.
3. Understand the protection schemes for different power system components.
4. Analyze model and simulation of protection schemes.
5. Understand the basic principles of digital protection.
6. Understand system protection schemes, and the use of wide-area measurements.

Text/References:

1. J. L. Blackburn, "Protective Relaying: Principles and Applications", Marcel Dekker, New York, 1987.
2. Y. G.Paithankar and S. R. Bhide, "Fundamentals of power system protection", Prentice Hall, India, 2010.
3. A. G. Phadke and J. S. Thorp, "Computer Relaying for Power Systems", John Wiley & Sons, 1988.
4. A. G. Phadke and J. S. Thorp, "Synchronized Phasor Measurements and their Applications", Springer, 2008.
5. D. Reimert, "Protective Relaying for Power Generation Systems", Taylor and Francis, 2006.

ELECTRICAL AND HYBRID VEHICLE

<table>
<tr><td colspan="5">Course Code</td><td></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electrical and Hybrid Vehicle</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. Acquaint the students to basic concepts of electric and hybrid electric vehicles.
2. To understand the power electronics devices and electrical machines.
3. To understand the different energy storage devices.
4. To understand different configurations of electric vehicles.

Unit-I

ELECTRIC VEHICLES
Introduction, Components, vehicle mechanics – Roadway fundamentals, vehicle kinetics, Dynamics of vehicle motion - Propulsion System Design.

Unit-II

BATTERY
Basics – Types, Parameters – Capacity, Discharge rate, State of charge, state of Discharge, Depth of Discharge, Technical characteristics, Battery pack Design, Properties of Batteries.

Unit-III

DC and AC ELECTRICAL MACHINES
Motor and Engine rating, Requirements, DC machines, three phase A/c machines, Induction machines, permanent magnet machines, switched reluctance machines.

Unit-IV

ELECTRIC VEHICLE DRIVE TRAIN
Transmission configuration, Components – gears, differential, clutch, brakes regenerative braking, motor sizing.
HYBRID ELECTRIC VEHICLES
Types – series, parallel and series, parallel configuration – Design – Drive train, sizing of components.

Course Outcomes:
At the end of the course, students will demonstrate the ability to:
1. Explain the basics of electric and hybrid electric vehicles.
2. Explain the architecture, technologies and fundamentals of EV.
3. Analyse the use of different power electronics devices and electrical machines in hybrid electric vehicles.
4. Explain the use of different energy storage devices used for hybrid electric vehicles, their technologies.
5. Explain the use of different control and select appropriate technology in EV.
6. Interpret working of different configurations of electric vehicles and its components, hybrid vehicle configuration, performance analysis and Energy Management strategies in HEVs.

Reference Books:

1. Iqbal Hussain, "Electric & Hybrid Vehicles – Design Fundamentals", Second Edition, CRC Press, 2011.
2. Mehrdad Ehsani, Yimin Gao, Ali Emadi, "Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals", CRC Press, 2010.
3. James Larminie, "Electric Vehicle Technology Explained", John Wiley & Sons, 2003.
4. Sandeep Dhameja, "Electric Vehicle Battery Systems", Newnes, 2000

MODELLING AND ANALYSIS OF ELECTRICAL MACHINES

Course Code					
Category	**Professional Elective Courses**				
Course title	**Modelling and Analysis of Electrical Machines**				
Scheme	**L**	**T**	**P**	**Credits**	**Semester: VI**
	3	**0**	**0**	**3**	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. To provide a comprehensive exposure to electrical machine design.
2. To understand various design principles and magnetic circuits.
3. To work on detailed designs.
4. To know about various computer aided design.

Unit-I

GENERAL: General features and limitations of electrical machine design. Types of enclosures, heat dissipation, temperature rise heating and cooling cycles and ratings of machine machines. Cooling media used.
BASIC DESIGN PRINCIPLES: Output equation and output coefficient, Specific electric and magnetic loading. Effect of size and ventilation.

Unit-II

MAGNETIC CIRCUITS: MMF calculation for airgun and iron parts of electrical machines, gap contraction coefficient. Real and apparent flux densities. Estimation of magnet current of transformers and rotating machines, no load current of transformers and induction motors. Leakage flux and reactance calculations for transformers and rotating machines, Design of field magnet.

Unit-III

DETAILED DESIGN: Design of transformer, D.C. machines induction motor and synchronous machine and their performance calculations.

Unit-IV

COMPUTER AIDED DESIGN: Computerization of design Procedures. Development of Computer program and performance prediction. Optimization techniques and their applications to design Problems.

Course Outcomes:
At the end of this course, students will demonstrate the ability to
1. Describe electrical machine design.
2. Analyze and apply various design principles.
3. Understand magnetic circuits.
4. Carry out detailed design.
5. Carry out proper computer aided design.
6. Apply optimization technique to design problems.

Text books:
1. A course in Electrical Machine Design by A.K. Sawhney, Khanna Pub.

Reference books:
1. Theory, performance and Design of alternating current machines by MG Say, ELBS, 15th Ed. 1986.
2. Theory, Performance and Design of Direct Current machines by A.E. Clayton, 3rd Ed. 1967. Optimization Techniques, S.S. Rao

ELECTRICAL SAFETY AND STANDARDS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Elective Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electrical Safety and Standards</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives:
1. To provide a comprehensive exposure to electrical hazards.
2. To understand various grounding techniques and safety procedures
3. To understand various safety audit Electrical safety programme structure
4. To know about various electrical maintenance techniques

Unit-I

Primary and secondary hazards- arc, blast, shocks-causes and effects-safety equipment- flash and thermal protection, head and eye protection-rubber insulating equipment, hot sticks, insulated tools, barriers and signs, safety tags, locking devices- voltage measuring instruments- proximity and contact testers-safety electrical one line diagram electrician's safety kit.

Unit-II

General requirements for grounding and bonding- definitions- grounding of electrical equipment bonding of electrically conducting materials and other equipment-connection of grounding and bonding equipment- system grounding- purpose of system grounding-grounding electrode system grounding conductor connection to electrodes-use of grounded circuit conductor for grounding equipment- grounding of low voltage and high voltage systems.

Unit-III

The six step safety methods- pre job briefings - hot-work decision tree-safe switching of power system- lockout-tag out- flash hazard calculation and approach distances- calculating the required level of arc protection-safety equipment, procedure for low, medium and high voltage systems- the one minute safety audit Electrical safety programme structure, development- company safety team safety policy programme implementation- employee electrical safety teams- safety meetings- safety audit accident prevention- first aid-rescue techniques-accident investigation

Unit-IV

Safety related case for electrical equipments, Various Standards: IEEE, IEC, IS..., regulatory bodies national electrical safety code-standard for electrical safety in work place- occupational safety and health administration standards, Indian Electricity Acts related to Electrical Safety.

Course Outcomes:

At the end of this course, students will demonstrate the ability to
1. Describe electrical hazards and safety equipment.
2. Analyze and apply various grounding and bonding techniques
3. Select appropriate safety method for low, medium and high voltage equipment.
4. Understand various safety audit Electrical safety programme structure
5. Participate in a safety team.
6. Carry out proper maintenance of electrical equipment by understanding various standards.

Text/References

1. John Cadick, Mary Capelli-Schellpfeffer, Dennis Neitzel, Al Winfield,'Electrical Safety Handbook', McGraw-Hill Education, 4thEdition, 2012.
2. Sunil S. Rao, Prof. H.L. Saluja, "Electrical safety, fire safety Engineering and safety management", Khanna Publishers. New Delhi, 1988.

3. Maxwell Adams.J, ‘Electrical Safety- a guide to the causes and prevention of electric hazards’, The Institution of Electric Engineers, IET 1994.
4. Ray A. Jones, Jane G. Jones, ‘Electrical Safety in the Workplace’, Jones & Bartlett Learning, 2000.

POWER SYSTEM II LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power System II Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

(i) At least 10 experiments are to be performed by students in the semester.

(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

(iii) Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS:

1. Draw the flow chart and develop the computer program for the formation of the Y Bus of a generalized network.
2. Draw the flow chart and develop the computer program for the formation of the Z Bus of a generalized network.
3. To plot the swing curve and observe the stability.
4. To perform load flow analysis using Gauss Seidel method.
5. To perform load flow analysis using Newton-Raphson method.
6. To study comparison of different load flow methods
7. To develop the program for stability analysis.
8. To observe transmission losses and efficiency with variations in power for the given example.
9. Simulation study on LFC of two area interconnected power system.
10. Simulation study on voltage control in multi area interconnected power system.

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes:

At the end of this lab, students will demonstrate the ability to;

1. Use numerical methods to analyse a power system in steady state.
2. Formulate Ybus and Zbus.
3. Apply load flow analysis on a power system.
4. Understand stability constraints in a synchronous grid.
5. Understand methods to control the voltage, frequency and power flow.
6. Understand the basics of power system economics

CONTROL SYSTEM LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Control System Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: V</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

(ii) At least 10 experiments are to be performed by students in the semester.

(iii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus. Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS: ANY SIX EXPERIEMENTS

1. To study speed Torque characteristics of
   a) A.C. servo motor
   b) DC servo motor.
2. (a) To demonstrate simple motor driven closed loop DC position control system.
   (b) To study and demonstrate simple closed loop speed control system.
3. To study the lead, lag, lead-lag compensators and to draw their magnitude and phase plots.
4. To study a stepper motor and to execute microprocessor or computer-based control of the same by changing number of steps, direction of rotation and speed.
5. To implement a PID controller for temperature control of a pilot plant.
6. To study behavior of 1st order, 2nd order type 0, type 1 system.
7. To study control action of light control device.
8. To study water level control using a industrial PLC.
9. To study motion control of a conveyor belt using an industrial PLC

SOFTWARE BASED (ANY FOUR EXPT.)

Introduction to SOFTWARE (Control System Toolbox)

10. Different Toolboxes in SOFTWARE, Introduction to Control Systems Toolbox.
11. Determine transpose, inverse values of given matrix.
12. Plot the pole-zero configuration in s-plane for the given transfer function. Plot unit step response of given transfer function and find peak overshoot, peak time.
13. Plot unit step response and to find rise time and delay time.
14. Plot locus of given transfer function, locate closed loop poles for different values of k.
15. Plot root locus of given transfer function and to find out S, Wd, Wn at given root and to discuss stability.
16. Plot bode plot of given transfer function and find gain and phase margins Plot the Nyquist plot for given transfer function and to discuss closed loop stability, gain and phase margin.

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes: At the end of this lab students will demonstrate the ability to

1. Understand the concepts of control systems and importance of feedback in control systems.
2. Perform signal flow graph and formulate transfer function.
3. Perform computations and solve problems on frequency response analysis.
4. Analyse Polar, Bode and Nyquist's plot.
5. Evaluate different types of state models and time functions.
6. Analyse different types of control systems like linear and non-linear control systems, etc.

MICROPROCESSORS AND MICROCONTROLLERS LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Microprocessors and Microcontrollers Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

(i) At least 10 experiments are to be performed by students in the semester.
(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
(iii) Group of students for practical should be 15 to 20 in number.

List of Experiments:

1. Write a program using 8085 and verify for:
   a) Addition of two 8-bit numbers.
   b) Addition of two 8-bit numbers (with carry).
2. Write a program using 8085 and verify for:
   a) 8-bit subtraction (display borrow)
   b) 16-bit subtraction (display borrow)
3. Write a program using 8085 for multiplication of two 8- bit numbers by repeated addition method. Check for minimum number of additions and test for typical data.
4. Write a program using 8085 for multiplication of two 8- bit numbers by bit rotation method and verify.
5. Write a program using 8086 for finding the square root of a given number and verify.
6. Write a program using 8086 for copying 12 bytes of data from source to destination and verify.
7. Write a program using 8086 and verify for:
   a) Finding the largest number from an array.
   b) Finding the smallest number from an array.
8. Write a program using 8086 for arranging an array of numbers in descending order and verify.
9. Write a program using 8086 for arranging an array of numbers in ascending order and verify.
10. Write a program to interface a two-digit number using seven-segment LEDs. Use 8085/8086 microprocessor and 8255 PPI.
11. Write a program to control the operation of stepper motor using 8085/8086 microprocessor and 8255 PPI.
12. To study implementation and interfacing of Display devices Like LCD, LED Bar graph and seven segment display with Microcontroller 8051/AT89C51
13. To study implementation and interfacing of Different motors like stepper motor, DC motor and servo Motors.
14. Write an ALP for temperature and pressure measurement
15. Write a program to interface a graphical LCD with 89C51

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes:

At the end of this lab students will be able to:

1. Understand the fundamentals of Microprocessors.
2. Understand the internal design of 8051 microcontroller along with the features and their programming.
3. Competent with the on-chip peripherals of microcontrollers.
4. Design different interfacing applications using microcontrollers and peripherals.
5. Demonstrate the limitations and strengths of different types of microcontrollers and their comparison.
6. Build systems using microcontrollers for real time applications.

PROJECT-I

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Project</td></tr>
<tr><td>Course title</td><td colspan="5">Project-I</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>2</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Course objectives:

1. To allow students to demonstrate skills learned during their course of study by asking them to deliver a product that has passed through the design, analysis, testing and evaluation
2. To encourage research through the integration learned in a number of courses.
3. To allow students to develop problem solving skills.
4. To encourage teamwork.
5. To improve students' communication skills by asking them to produce both a professional report and to give an oral presentation and prepare a technical report.

The students are required to undertake institutional project work.

The final Viva voice of the institutional project work will be conducted by an external examiner and one external examiner appointed by the institute. External examiner will be from the panel of examiners submitted by the concerned institute approved by the board of studies in engineering and technology. Assessment of institutional project work will be based on seminar, viva-voice and report of institutional project work obtained by the student from the industry or institute.

The internal marks distribution for the students consists of 50 marks internally and 50 marks by an external examiner.

Course outcomes

On successful completion of the course students will be able to:

1. Demonstrate a sound technical knowledge of their selected project topic.
2. Undertake problem identification and formulation.
3. Design engineering formula to complex problems utilising a systems approach.
4. Research and engineering project.
5. Communicate with engineers and the community at large in written and oral form.
6. Demonstrate the knowledge, skills and attitudes of a professional engineer.

ECONOMICS FOR ENGINEERS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Non-Credit</td></tr>
<tr><td>Course title</td><td colspan="5">Economics for Engineers</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: VI</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Class Work</td><td colspan="5">30</td></tr>
<tr><td>Exam</td><td colspan="5">70</td></tr>
<tr><td>Total</td><td colspan="5">100</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3 Hrs</td></tr>
</table>

Note: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

Course Objectives:

1. Acquaint the students to basic concepts of economics and their operational significance.
2. Acquaint students with market and its operation.
3. To stimulate the students to think systematically and objectively about contemporary economic problems.

Unit-I

Definition of Economics- Various definitions, types of economics- Micro and Macro Economics, nature of economic problem, Production Possibility Curve, Economic laws and their nature, Relationship between Science, Engineering, Technology and Economic Development.

Demand- Meaning of Demand, Law of Demand, Elasticity of Demand- meaning, factors effecting it, its practical application and importance.

Unit-II

Production- Meaning of Production and factors of production, Law of variable proportions, Returns to scale, Internal and external economies and diseconomies of scale.

Various concepts of cost of production- Fixed cost, Variable cost, Money cost, Real cost, accounting cost, Marginal cost, Opportunity cost. Shape of Average cost, Marginal cost, Total cost etc. in short run and long run.

Unit-III

Market- Meaning of Market, Types of Market- Perfect Competition, Monopoly, Monopolistic Competition and Oligopoly (main features).

Supply- Supply and law of supply, Role of demand and supply in price determination and effect of changes in demand and supply on prices.

Unit-IV

Indian Economy- Nature and characteristics of Indian economy as under developed, developing and mixed economy (brief and elementary introduction), Privatization - meaning, merits and demerits. Globalization of Indian economy - merits and demerits. Banking- Concept of a Bank, Commercial Bank- functions, Central Bank- functions, Difference between Commercial and Central Bank.

Course outcomes:

1. The students will able to understand the basic concept of economics.
2. The students will able to understand the basic concept of demand.
3. The student will able to understand the concept of production and cost.
4. The student will able to understand the concept of market.
5. The students will able to understand the basic concept of supply.
6. The student will able to understand the concept of privatization, globalization and banks.

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